

# Karan Gupta

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## EDUCATION

### University of Waterloo

*Bachelor of Applied Science in Computer Engineering, Minor in Economics*

Sep. 2025 – Present

**GPA: 3.95**

## EXPERIENCE

### Software Development Engineer Intern

*Amazon*

May 2026 – Present

*Toronto, ON*

- Developing an agentic AI refactoring workflow to migrate legacy optimization code into a modernized framework using RAG for context-aware code translation.
- Developing multi-step AI agents to automate syntax transformation and identify logic bottlenecks,
- Engineering a natural language interface to enable on-the-fly algorithmic adjustments and rule modifications during the automated migration.

### Software Engineering Intern

*Manulife Financial Corporation*

January 2026 – April 2026

*Waterloo, ON*

- Accomplished proactive system reliability by scripting a New Relic data exporter and architecting NRQL queries, resulting in the **early detection of 3+** resource bottlenecks before system failure.
- Engineered a Python analysis script and SQL telemetry pipeline to process clickstream data, **identifying 20+** suspicious user anomalies by cross-referencing individual user IDs with action-based behavioral logs.
- Delivered **15+** feature deployments using Apex and LWC, maintaining an **87%** test coverage target through **Copado** and **Jira** for project tracking.

### Autonomy Software Developer

*Waterloo Aerial Robotics Group*

Sep. 2025 – Present

*Waterloo, ON*

- Implemented computer vision algorithms using OpenCV2 for **real-time signal detection**, improving recognition accuracy through iterative parameter tuning and validation.
- Built multi-process telemetry and command systems in Python (PyMAVLink) to simulate UAV communication, enhancing reliability and throughput across distributed processes.

## PROJECTS

### DeliriumWatch | *Raspberry Pi, Arduino, Python, OpenCV, HTML, CSS, Flask, C/C++* Sep. 2025 – Nov. 2025

- Built a real time Python monitoring pipeline with SQL-backed secure login and role-based access control for hospital staff, resulting in over **90% reduction** in manual environmental monitoring during simulated testing.
- Implemented OpenCV based eye detection and live Flask web visualizations to flag abnormal conditions during clinical validation.

### Reparo (Hack Canada 2026 Winner) | *Python, Gemini API, React, Node.js, Swift* Mar 2026 – Present

- Won **1st place** (\$5,000) by building an **Agentic AI** system using Gemini vision models for product classification and reparability assessment with **90%+ accuracy**.
- Reduced user search time for replacement parts by **70%** and enabled access to **1,000+ real-time listings** by integrating Shopify Storefront API and SerpAPI to automate matching and streamline repair-to-checkout flow
- Won **1st place** in Reactiv Track (**\$5,000**) at Hack Canada 2026; shortlisted for **Most Complex AI Hack**

### TailorAI (Sillhouette) | *Python, OpenCV, Node.js, React, MongoDB* June 2025 – Present

- Built a full-stack AI platform using computer vision and the MERN stack, processing **200+ images** to automatically extract **10+ body measurements** per user and eliminating reliance on manual sizing inputs
- Implemented Python CV pipeline with landmark normalization, improving measurement consistency by **40%**
- Designed scalable backend APIs, MongoDB schema, and responsive React frontend, supporting **150+** item profiles, **sub-200 ms query latency**, and ongoing development in collaboration with venture partners

## TECHNICAL SKILLS

**Languages:** Python, Java, C, C++, TypeScript, JavaScript, Node.js, SQL, Apex, HTML/CSS, VBA

**Frameworks & Libraries:** Retrieval-Augmented Generation (RAG), React, Node.js, Flask, FastAPI, OpenCV2, REST API, PyMavLink, NumPy, JUnit

**Developer Tools:** Git, GitHub, Docker, VS Code, AWS (S3, SQS, SNS), Salesforce, Postman

**Cloud and Databases:** Relational Databases (SQL), MongoDB, Familiarity with Cloud Technologies